

Mac-2-DOSTM

*File Transfer Utility
For the Amiga[®] Computer*

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Overview

Thanks for purchasing Mac-2-DOS. All of us at Central Coast Software appreciate your support, and we promise to do our best to provide high quality programs for your Amiga.

Mac-2-DOS makes it easy and simple for you to transfer text and data files on diskette between your Amiga and a Macintosh. You probably already know that normally the Amiga cannot read Mac disks and the Mac cannot read Amiga disks. We could speculate forever about the corporate decisions which led to this situation, but Mac-2-DOS now provides a simple solution: attach a Macintosh floppy disk drive to the Amiga and teach the Amiga to use it!

Mac-2-DOS transfers Macintosh files to and from the Amiga using a standard Macintosh floppy disk drive attached to the external floppy disk drive connector of the Amiga. Mac-2-DOS includes a small “interface” adapter which lets you connect the Mac drive directly to your Amiga. Mac-2-DOS includes software which understands the Mac file structure and converts Mac file formats to forms which are usable on your Amiga.

Note: Mac-2-DOS is *not* a Macintosh emulator. It *does not* let you run Mac programs on your Amiga. Mac-2-DOS is not a communications program. It *requires* a Macintosh floppy disk drive.

Hardware and Software Requirements

Mac-2-DOS works with any standard Amiga equipped with at least 256K of memory. It works fine with accelerator boards and 68020s. It was designed to handle overscan, NTSC (US) and PAL (European) monitors. It has no problems on machines equipped with the Fat AGNUS chip.

Mac-2-DOS requires a standard Macintosh-compatible 3.5-inch external floppy disk drive. Mac-2-DOS works with either single-sided (400k) or double-sided (800k) Mac drives. It has been tested with all available third-party Mac drives. The Mac drive **MUST** be connected directly to the Mac-2-DOS interface. Mac-2-DOS cannot use Amiga floppy disk drives to read/write Macintosh diskettes.

Mac-2-DOS requires Workbench V1.2 or later, and works fine with the fast file system.

The Amiga was designed to use up to three external floppy drives (the A500 is restricted by power supply limitations to *one* external floppy drive). Mac-2-DOS connects a Mac external floppy drive to the Amiga. The Mac drive draws its power from the Amiga and counts as one of the maximum of three external drives. Mac-2-DOS must be the **LAST** device in the floppy drive daisy-chain. This is because it does not have a daisy-chain connector for additional drives.

Mac-2-DOS works with other, nonstandard devices connected in the floppy drive chain, provided those devices pass through all standard floppy signals.

Mac-2-DOS is available in two versions: Package A, intended for those who already have an existing Macintosh drive and do not need or want to purchase another Mac drive, and Package B, which includes all features of Package A and adds a Macintosh-compatible floppy drive which functions both as a Mac drive and a normal Amiga drive.

Note: Most Mac-compatible floppy drives, including drives from Apple, Cutting Edge, and Mirror, *cannot* function as an Amiga drive. The Mac-compatible drive provided by CCS as part of Package B is the only floppy drive which can perform as an Amiga drive. If you want your Mac drive to work as an Amiga drive, you *must* purchase Package B, which includes this Mac drive.

Hardware Installation

WARNING — WARNING — WARNING

DO NOT CONNECT THE MAC-2-DOS INTERFACE TO YOUR AMIGA WITH THE POWER TURNED ON. ALWAYS TURN OFF THE POWER BEFORE CONNECTING OR DISCONNECTING THE INTERFACE OR ANY FLOPPY DRIVES. FAILURE TO FOLLOW THIS SIMPLE RULE MAY RESULT IN DAMAGE TO YOUR AMIGA AND/OR THE MAC-2-DOS INTERFACE.

The Mac-2-DOS interface adapter plugs into the external floppy disk drive connector on the back of your Amiga or into the last external Amiga floppy drive, if any. The Mac-2-DOS interface **MUST** be the last device in the chain of floppy drives.

Follow these steps to install the Mac-2-DOS interface:

1. Turn off the Amiga power. **DO NOT** connect the interface with the power turned on.
2. Locate the external floppy drive connector on the back of your Amiga. This is a 23-pin female connector. If there is already an external Amiga floppy drive connected here, find the last floppy drive of the chain and locate its 23-pin female connector.
3. Carefully insert the 23-pin male end of the Mac-2-DOS interface into the external floppy drive connector.
4. Connect a Macintosh-compatible 3.5-inch external floppy disk drive to the 19-pin female end of the Mac-2-DOS interface.
5. Turn on the Amiga power.

If you already have three Amiga external floppy drives connected to your computer, you must remove one of them so that you can connect Mac-2-DOS. You cannot exceed three external floppy drives. Mac-2-DOS cannot function as a fourth or later external device.

Attention A500 owners: the A500 power supply is not designed to handle more than one external floppy drive. Most A500s seem to be able to operate with two external drives, but there is always the chance your machine may have a “weak” power supply. Because of the chance of damage to your computer, we cannot recommend using Mac-2-DOS on an A500 with another external drive. We suggest you remove the other external drive before connecting Mac-2-DOS.

Commodore attempted to enforce the limit of one external floppy on the A500 by selling “modified” A1010 external floppy drives *which will not drive* another external floppy drive, even though there is a daisy-chain connector on the back of the drive. If you have one of these drives, you *will not* be able to operate Mac-2-DOS as the second external drive. If Mac-2-DOS fails to function properly as the second external drive on an A500, we suggest that you remove the Amiga external floppy drive and connect Mac-2-DOS directly to the A500.

Software Installation

The Mac-2-DOS disk is not copy protected. We encourage you to make a backup copy of the Mac-2-DOS master disk for your own protection. And be sure to put the master disk in a safe place. You will be asked to return the master disk to us for any upgrades we may offer.

Mac-2-DOS is available in two versions: Package A, which consists of the interface adapter, Mac-2-DOS software, and a manual; and Package B, which includes all of the above plus a modified Macintosh-compatible floppy disk drive and special driver software. If a floppy drive with a Mac-2-DOS label on top was included with the product when you purchased Mac-2-DOS you have Package B. Otherwise you have Package A. Caution: some Amiga dealers are selling Mac-2-DOS Package A bundled with a third-party Mac floppy drive. Such drives *do not* have a Mac-2-DOS label on the top of the drive. For purposes of installation, this combination is treated as Package A.

Package A software installation is quite simple: copy the Mac-2-DOS program into the drawer where you wish to use it or into the C directory of your favorite Workbench disk or onto your hard disk. From Workbench drag the Mac-2-DOS program icon (*not* the disk icon!) to the desired drawer, or from CLI use the “Copy” command to copy the program and its icon, as you prefer. That’s all there is to it!

Package B software installation is somewhat more complicated. The Mac-2-DOS Package B diskette contains two drawers, labeled Package A and Package B. If you prefer, you can install the Package A software. This is simple and fast, and gets you going with little hassle. Open the Package A drawer and copy the Mac-2-DOS program to wherever you want it, as described above. However, Package A *does not* permit the use of the Mac-2-DOS drive as an Amiga drive.

To use your modified Mac-2-DOS floppy drive as both a Mac drive and an Amiga drive, you must install the Package B software, which includes a special AmigaDOS device driver that is not part of Package A.

Here is what to do:

1. Click on the icon for InstallM2D. This program copies the Mac-2-DOS program and icon to the root directory of your SYS: device, adds a new device named MF0: to your MountList and adds the command

MOUNT MF0:

to your startup-sequence.

That's all that is required. Your new AmigaDOS device is known as MF0:, and can be used as a normal Amiga floppy for reading and writing standard Amiga floppy disks.

As an alternative to the “automatic” InstallM2D installation procedure, you may wish to perform the installation manually. Here are the steps:

1. Copy the Mac-2-DOS program from the drawer named Package B to the drawer where you want to keep it. Use the CLI COPY command or drag the icon, as you prefer.
2. Copy the MacFloppy.device driver from the drawer named Package B to the DEVS: directory on your Workbench disk.
3. Using your favorite text editor, such as ED, EDIT, or TxED, edit the MountList file on your Amiga to include the new floppy device. Then edit the Startup-sequence to Mount the new device when you boot your Amiga. The necessary MountList information is provided in Appendix A. Mac-2-DOS expects the device to be named MF0:, and does not recognize any other device name. Note that the flags parameter *must* be 1, as shown in Appendix A.
4. Type: Mount MF0:

That completes the installation. Start Mac-2-DOS, as described in the following section, and enjoy!

Getting Started

Ready to transfer some Mac files? Make sure you have installed the Mac-2-DOS interface and software, as described above. If not, do it right now.

To start Mac-2-DOS:

- From Workbench, double-click on the Mac-2-DOS icon.
- From CLI, type Mac-2-DOS.

When Mac-2-DOS first starts up, it attempts to read the floppy in the Mac drive (if any) and displays the contents of the root directory from that floppy in the left directory window. Then it reads the current AmigaDOS directory and displays the contents in the right directory window.

If no Mac disk is loaded, you should load one now. If you wish to transfer Amiga files to a Mac, you must have a disk formatted as a Mac disk. You cannot transfer the files to a Mac on an Amiga disk. If you have no Mac disk, but wish to create one, see the section entitled **Format Disk**.

To copy a file from one side to the other, follow these steps:

1. First, click on the name of the file you wish to transfer. Notice that the color of the file name changes, and the orange COPY arrow at the upper center of the display points to the opposite directory window. That arrow indicates the direction of the transfer.
2. Next, select the proper file conversion from the Conversion menu. The proper file conversion depends entirely on the type of file you wish to transfer. This subject is discussed in detail in About Mac Files, About Amiga Files, and Mac-2-DOS File Conversion Options.
3. Next, you may be asked to enter either the Mac file type and creator codes (if transferring to the Mac) or the Amiga default tool name (if transferring to the Amiga). See **Mac-2-DOS File Conversion Options**.
4. Finally, click on the COPY arrow to start the file transfer process. The Copy Status screen displays the progress of the file transfer.
5. When the file transfer is complete, look in the opposite directory window to see that the file has been successfully transferred to the other side.

When you have finished transferring files, click on the Close gadget or choose Quit from the Project menu. Mac-2-DOS ejects the Mac disk, if any, from the Mac drive as it exits.

Selecting Files and Directories

The directory windows display the names and sizes of files and the names of subdirectories (folders), sorted alphabetically. Subdirectories are listed first, with files following. Use the scrolling gadgets at the right of the window to move up or down in the list to view all of the names if there are more names than can be displayed at once.

To select a file for transfer and set the direction of the transfer (from Mac to Amiga or from Amiga to Mac), simply click anywhere on the name of the file you wish to transfer. The file name changes color to indicate its selection. Only files which have been selected in this manner are transferred by Mac-2-DOS.

To deselect a file after it has been selected, click anywhere on the file name. The color changes back to its original color. Note that when you select the first file in one window, the COPY arrow changes direction to indicate that the direction of copy is toward the other window, and any files that were selected in the other window are automatically deselected. Mac-2-DOS does not support copying files in both directions in one step.

You may select as many files as you wish within a given directory. All selected files are transferred, one at a time, in one COPY operation, but the same file conversion is applied to all selected files. If you wish to transfer files requiring different file conversions, you must group the files by type of file conversion and perform separate COPY steps for each group of files.

You can use the Select Files menu to quickly select or deselect all files in a directory or select or deselect groups of files by name with wildcards. Mac-2-DOS uses the characters # and ? in the usual Amiga way to indicate wildcard options when selecting files by name.

For example, if you wish to transfer a certain group of files having different names, but all ending in .PIC, you would enter the file name as:

#?.PIC

Mac-2-DOS then selects for transfer all files with a name ending in .PIC in that directory.

If you wish to change to a subdirectory (folder), simply click on the name of the subdirectory (folder). To change to a parent or root directory, click on the parent or root directory gadget below the directory window. Note: when you change to a different directory, all files which you have selected in the directory you are leaving are automatically deselected. File transfer operations only apply to the directory currently displayed in each window.

If you wish to change Mac disks, you can eject the disk from the Mac drive by clicking on the Mac disk volume name or selecting Eject Disk from the Mac Disk menu.

If you wish to change to a different Amiga volume or to change to a different path on a volume, click on the Amiga volume name, and then enter the device:path name you wish to use. Mac-2-DOS changes to that directory and displays the files available for transfer. This process deselects all Amiga files which you may already have selected for transfer.

About File Names

Fortunately, the Mac and the Amiga use similar naming conventions for their files. The Mac allows up to 31 characters in a file name. The name cannot contain a colon (:) and shouldn't start with a period (.). All other characters on the keyboard are acceptable in a Mac file name. The Amiga allows up to 30 characters in a file name. The name cannot contain a colon (:) or slash (/). All other characters on the keyboard are acceptable in an Amiga file name.

In almost all cases Mac-2-DOS preserves the original name of the file when you transfer a file from one machine to the other. However, if the Mac file name is exactly 31 characters, Mac-2-DOS drops the last character when it transfers the file to the Amiga. Likewise, if a Mac file name contains a slash (/) or an Amiga file name contains a leading period (.), Mac-2-DOS deletes the “illegal” character from the file name when it transfers the file.

About Macintosh Files

The following discussion is a bit technical, but some of this is necessary to understand the problems you face in using an Amiga file on a Mac after you have copied it to a Mac disk.

A Mac file consists of two parts: the “data fork”, which holds what we normally think of as the data of a file, and the “resource fork”, which holds special information, such as the icon, fonts, etc., which may apply to that particular file. In many files the resource fork is empty. In any case, the content of the resource fork is highly specific to the type of Mac file, and is generally useless to the Amiga. The Amiga has no equivalent of the resource fork. For that reason, Mac-2-DOS transfers only the data fork, except in one specific case.

The resource fork contains the executable code for Mac programs. Mac programs *cannot* run on the Amiga and Amiga programs *cannot* run on the Mac. So you might think there is no value in transferring them. However, Mac-2-DOS supports the MacBinary method of transferring both the resource and data forks of a Mac program. When a Mac program is transferred in MacBinary format, it can be successfully transported over modems and via non-Mac computers, such as the Amiga.

Mac files are identified to the Mac by two 4-character fields in the file header. The “file type” field identifies the general content of the file, while the “file creator” field identifies the Mac program which created the file and is supposed to process that file. A Mac program called “Finder” manages the Mac screen (called the “Desktop”). Finder is functionally equivalent to the Amiga program “Workbench”. When you double-click on a Mac file icon, Finder looks at the 4-character file creator code to decide which Mac program is supposed to process that Mac file, then starts up that program and passes the name of the file you selected to that program. The file creator code performs the same function on a Mac as the “default tool” field of the “.info” file performs on an Amiga.

If you plan to transfer an Amiga file to the Mac and then be able to click on the file icon to start up the proper Mac program, as you would on the Amiga, you must make sure that Mac-2-DOS sets the file type and file creator codes correctly for the Mac program which you expect to process the file. Appendix B lists file type and creator combinations for many common Mac programs.

In some cases Mac-2-DOS knows what file type and creator to assign to a new Mac file based on the type of conversion you select. In other cases you must tell it what to put into these fields. For example, if the conversion you specify is IFF to MacPaint, Mac-2-DOS knows to make the file type PNTG and the creator MPNT. For ASCII text files and PostScript files, Mac-2-DOS assigns the file type to be TEXT and the creator MACA (MacWrite). For MacBinary files, the file type and creator are set according to information contained in a special

header of the MacBinary file.

However, for No Conversion files, where Mac-2-DOS does not assume anything about the content of the file, you must provide the necessary file type and creator information. If these fields are not set correctly, you may have problems using your files on a Mac.

Appendix B may help you determine how to set these fields. Another method of finding the proper file type and creator is to use the File Info option of the Mac-2-DOS File Utilities menu to examine an existing file of that type created on the Mac by the desired program. You may also be able to get the necessary information from the company which produced the Mac program.

Mac files *always* have an icon. Even if a file has no custom icon defined for it, the Finder assigns the file a standard generic icon. For that reason, Mac-2-DOS does not assign a custom icon to a file, but lets Finder provide a generic icon. Of course, after you process a file using a Mac program, that program will usually assign its own custom icon to the file.

About Amiga Files

The following discussion is a bit technical, but some of this is necessary to understand the problems you face in using a Mac file on an Amiga after you have copied it from a Mac disk.

Amiga files do not have a “data fork” or “resource fork” as do Mac files. However, many Amiga files have a supplemental file with a similar name which contains information about the main file. This supplemental file contains the file icon and name of the Amiga program which is supposed to process the file. The supplemental file can easily be identified, because it has the same name as the main file, with the addition of the characters “.info” at the end.

For example, if the main file is named:

Ra lph

the supplemental file containing Ralph’s icon is called:

Ra lph . i n f o

The “.info” file is used by Workbench to display the icon of the file. In fact, if a file has no supplemental “.info” file, no icon will appear on the Workbench screen for that file. There is no standard, generic icon for Amiga files which have no “.info” file. Such files are effectively invisible to Workbench, and can only be selected or processed from CLI or from inside an Amiga program using the Open File option (if any).

The supplemental “.info” file serves another function: it defines the name of the program which created the file and should be invoked to process the file. This information is called the “default tool”. The default tool field in the “.info” file performs the same function on the Amiga as the 4-character file creator code on the Mac.

For example, if you use WordPerfect to create a document called Bill, WordPerfect will also create a supplemental file named Bill.info containing the icon for Bill with the default tool field set to “WordPerfect”. This way when you want to edit document Bill, all you have to do is double-click on Bill. Workbench invokes WordPerfect and tells it to process Bill.

Remember, Amiga files do not necessarily have an icon. Some Amiga users do not want the loss of disk space or clutter of the “.info” files on their Amiga disk. However, remember that if a file has no “.info” file, you will *not* be able to process the file directly from Workbench.

When you transfer Mac files to the Amiga, you can decide whether or not you want Mac-2-DOS to create the supplemental “.info” file for each Mac file you transfer to the Amiga. To suppress the “.info” files, deselect the Amiga Icon item on the Other Options menu.

If you specify that Mac-2-DOS is to create an “.info” file, you must tell Mac-2-DOS the name of the program (default tool) you want to process that file. IF you do not provide the name, Workbench will be unable to invoke the program when you double-click on the file’s icon. If you enter the wrong program name or no name at all for the default tool, you can always change it at a later time using the Info item from the Workbench menu.

If you do not know what default tool to assign to a Mac file which you wish to transfer to the Amiga, use the info item from the Workbench menu to examine an existing Amiga file produced by that program.

Some Amiga programs are very specific about the names of files which they will accept as input. For example, VIP Professional expects the file name to end in “.wks”. In this case you may have to rename the file (and possibly its “.info” file) from its original Mac name to make the file acceptable to the Amiga program. You can easily rename a file using the Rename item from the Workbench menu.

Mac-2-DOS Conversion Options

Mac-2-DOS supports five file conversion options. When you transfer a group of files, the one conversion option you have selected is automatically applied to all files of the group. If you wish to apply different conversions to different files, the files must be transferred separately, not as a group.

Each of the Mac-2-DOS file conversion options (with the exception of MacBinary) uses “standard” file type and creator codes when transferring a file to the Mac and a “standard” default tool when transferring a file to the Amiga. You can change the file type and creator codes and default tool for each conversion option to tailor the conversion to your particular requirements. To edit these parameters, select the Edit Types/Tools menu item from the Project menu. Then to make these changes permanent, select Save Options from the Project menu.

The following is an explanation of each of the conversion choices, and the type of files to which it applies.

No Conversion

This option transfers a Mac data file to or from the Amiga with no changes to the content or length of the file. This option is most useful when you know that the Amiga and Mac programs you plan to use share a common file format, and the file you wish to transfer has been stored in that compatible format.

For example, you would use this option to transfer a document produced by WordPerfect on the Amiga for use with WordPerfect on the Mac. Similarly, you would use this option to transfer a spreadsheet produced by Microsoft Excel on the Mac to use with Analyze on the Amiga. In this latter case, the compatible file format is Lotus 123, which both spreadsheet programs can process. You would “export” the spreadsheet in “.wks” format using Excel, and then “import” the “.wks” file into Analyze after transferring it to the Amiga using Mac-2-DOS.

See the section entitled **Mac/Amiga File Compatibility** for further discussion of Mac/Amiga file compatibility.

For files transferred to the Mac using No Conversion, Mac-2-DOS must know what file type and creator codes to assign to the file. See Appendix B for a list of common Mac file type and creator codes. Also see the section entitled **About Mac Files**.

For files transferred to the Amiga using No Conversion, Mac-2-DOS must know what default tool to assign to the file if the file is to be given an icon. See the section entitled **About Amiga Files**.

MacBinary

This option transfers both the data fork and the resource fork of a Mac program file in a special format which preserves all information necessary to permit the program to be run successfully on a Mac.

For example, the Mac program ResEdit is stored on the Mac-2-DOS master diskette as an Amiga file containing data in MacBinary format, so that you can transfer it to your Mac, if you wish. Once on the Mac disk, the program can be run just as if it had originated there, when, in fact, it came to you on an Amiga disk as an Amiga file.

Mac file type and creator codes are contained within the MacBinary file format. When transferring MacBinary files, Mac-2-DOS does not ask for these codes.

Note: This option is *not* useful for transferring Amiga programs to the Mac.

ASCII (text)

This option transfers text files in ASCII format between the Mac and the Amiga. This is most useful for transferring text and document files between the two machines.

The only word processing programs which offer direct file compatibility between the Mac and the Amiga (as of July, 1989) are WordPerfect Amiga and WordPerfect Mac. For all other cases where you wish to transfer a document between incompatible word processing programs on the two machines, you must save the document as an ASCII file (pure text, no formatting), transfer it to the other machine, and then import it as an ASCII file using the other word processor.

The ASCII (text) option automatically converts line ending characters and translates the extended international character sets, which differ on the two machines.

Note that some programs, such as MacWrite, place line ending characters only at the end of a paragraph. If there is a choice of ASCII formats in your word processing program, for the best compatibility you should choose the option which places line ending characters at the end of each line.

When transferring ASCII files to the Mac, Mac-2-DOS assigns a file type of TEXT and creator of MACA (MacWrite) to these files. When transferring these files to the Amiga, Mac-2-DOS assigns the default tool NotePad to these files.

PostScript

This option is used to transfer PostScript files between the two machines. Note that although a PostScript file is actually an ASCII text file, the Encapsulated PostScript format can contain graphic images in bitmapped form. For this reason

Mac-2-DOS does *not* apply the normal ASCII conversion to these files, which could corrupt the bitmap portion of the file.

When transferring PostScript files to the Mac, Mac-2-DOS assigns a file type of TEXT and creator of MACA (MacWrite) to these files. When transferring these files to the Amiga, Mac-2-DOS assigns the default tool NotePad to these files.

MacPaint-IFF

This conversion option converts MacPaint bitmapped image files to/from IFF InterLeavedBitMap (ILBM) form. This can be very useful for transferring clip art images to the Amiga from the Mac.

MacPaint images are always stored in black and white, with dimensions of 576x720 pixels. IFF ILBM images can be up to 32 colors, with pixel dimensions determined by the image size in the IFF file. Mac-2-DOS supports IFF HAM, half-brite, and overscan modes as well.

When Mac-2-DOS converts a MacPaint image to IFF, it sets the image size to 640x720 pixels, and the Amiga screen resolution as determined by the user. The default tool is set to Deluxe Paint. Note that the resulting image is always black and white (one bitplane), and is a direct (pixel by pixel) translation of the original MacPaint image.

When Mac-2-DOS converts an IFF image to MacPaint, it must combine the multiple (color) bitplanes into a single (black and white) bitplane. To accomplish this conversion, Mac-2-DOS applies an ordered dithering algorithm to the ILBM data, similar to that performed by the Amiga when a color ILBM image is printed on a black and white printer. The resulting image is clipped to 576x720 pixels. Mac-2-DOS sets the file type to PNTG and the creator to MPNT (MacPaint). Note that this conversion process is *very* slow compared to the other conversions, and the resulting MacPaint file is usually much smaller than the original IFF file.

The MacPaint image is limited to 576 pixels horizontally by 720 pixels vertically. If the IFF image greatly exceeds these limits, some of the original image will be lost when the image is clipped during the conversion. To minimize the impact of this clipping, you may wish to position the IFF image as far as possible toward the upper left corner of the Amiga screen before converting the image to MacPaint format.

Mac/Amiga File Compatibility

It's one thing to physically transfer a file to or from a Mac-format disk. Mac-2-DOS does that job well. But finding a way to use that information once it is on the destination machine is another problem entirely. This section explains some of the issues of file compatibility between the Macintosh and the Amiga and should help you to develop conversion strategies to get your data "from here to there" in a usable form.

This section includes the following file categories:

- Word Processing
- Desktop Publishing
- Spreadsheets
- Databases
- Graphic/Art/Paint
- Computer Assisted Design
- Music
- Clip art
- Fonts

This section concludes with a list of various Mac and Amiga file conversion utility programs which we think might be useful to you.

Word Processing

There are two ways to transfer word processing documents between the Mac and the Amiga: (1) as ASCII text files, or (2) between word processing programs which store documents in an identical format. Mac-2-DOS *does not* convert formatted Amiga documents into formatted Mac documents or vice versa. The reason is that there are just too many different proprietary document formats used by the various word processing programs for us provide document format conversion. Therefore, your documents must be stored in some compatible format, such as ASCII text, in order to be usable after they are transferred to the other machine using Mac-2-DOS.

Currently the only word processing program which is available for both the Mac and the Amiga is WordPerfect, and even in this case the files are not totally compatible between these two versions of WordPerfect.

When using WordPerfect on the Mac, first save the document as a V4.2 document. Then transfer the document to the Amiga with Mac-2-DOS, using the No Conversion option. Once the document is on the Amiga, you must convert it from V4.2 format to V4.1 format using a conversion utility which is supplied by WordPerfect Corporation. Contact WordPerfect Corp. for more details.

Using WordPerfect V4.1 on the Amiga, just save the document in its normal format. Then transfer the document to the Mac with Mac-2-DOS, using the “No Conversion” option. No other steps are required.

To transfer documents between all other word processing programs on the two machines, our research reveals that the you *must* use the ASCII text mode. First save the document as a text file. Usually this is *not* the normal mode for storing documents. If your word processing program offers a choice of placing line endings at the end of every line or at the end of a paragraph, you will probably want to choose “at the end of every line” for the best compatibility. Then transfer the document to the other machine with Mac-2-DOS, using the ASCII (text) conversion. Finally, “import” the document into the other word processing program as an ASCII text file. You may lose much of the special formatting, such as footnotes, but this technique does preserve the text of the document.

Most, if not all word processing programs can import and export documents in ASCII format. Often, this format is called “Text Only” format.

Desktop Publishing

Most desktop publishing programs store their documents in a proprietary format. That means that it is usually quite difficult to transfer page layout files between such programs, even on the same machine. However, almost without exception, desktop publishing programs produce printer output files in PostScript (PS) format. Usually, these programs accept input documents in text form and often accept graphic images and clip art in bitmap form. So it quite reasonable to generate text and graphic images on one machine and transfer those to the other machine where they can be easily used by a page layout program.

In the world of desktop publishing (DTP), PostScript (PS) is the standard output file format supported by the majority of programs on both the Mac and the Amiga. Most programs that do not support PostScript are being revised to include this capability. Mac-2-DOS provides a PostScript conversion option which is used to transfer PostScript files in either direction.

A form of PostScript called Encapsulated PostScript (EPS) has been defined which includes a bitmap image “encapsulated” within the PostScript file. Unfortunately, there is no universal standard for this bitmap image. In practice, the image is stored in a form compatible with the machine on which the file is generated. This means that Mac EPS files contain a bitmap image which is not compatible with IBM EPS files, and vice versa. Currently there is no EPS bitmap image standard for the Amiga, although IFF ILBM format would seem appropriate.

The Mac-2-DOS PostScript conversion option preserves but *does not convert* any bitmap image present in an EPS file.

To print a PostScript file on a Mac after the file has been transferred from an Amiga to a Mac disk, simply click once on the file icon to “select” the file, then select Print from the File menu. This sends the file to the printer port of the Mac. Obviously, for a PostScript file this is useful only if the Mac printer is a LaserWriter or other PostScript-compatible printer. Note that no special program is needed to print a PostScript file on the Mac. To print a PostScript file on the Amiga after the file has been transferred to the Amiga, you must use a print utility program or load the file into a desktop publishing program, such as Professional Page.

Spreadsheets

WKS (Lotus), SYLK (Microsoft) and DIF (Microsoft) are the standard file formats for spreadsheet files, supported by many programs on both the Mac and the Amiga. Many spreadsheets also can import and/or export data in ASCII format.

To transfer spreadsheet files, first determine if the spreadsheet programs you plan to use on the two machines share a common standard format, such as WKS, SYLK or DIF. If so, store the spreadsheet file in that format, transfer it with Mac-2-DOS using the No Conversion option, and load it into the destination program.

If the programs do not share a common spreadsheet file format, you may be able to transfer the file as an ASCII (text) file. Find out if both spreadsheets offer an ASCII “export/import” option. If so, “export” the file in ASCII, transfer with Mac-2-DOS using the ASCII (text) conversion option, and finally “import” the file in ASCII format on the destination machine.

Databases

The dBASE format has long been a de facto standard for database files. Many database programs for both the Mac and the Amiga support dBase files. Most database programs also import/export ASCII files, with fields delimited with tab, comma, or quotes.

To transfer database files, first determine if the database programs you plan to use on the two machines share a common standard format, such as dBase. If so, store the database file in that format, transfer it with Mac-2-DOS using the No Conversion option, and load it into the destination program.

If the programs do not share a common database file format, you may be able to transfer the file as an ASCII (text) file. Find out if both spreadsheets offer an ASCII “export/import” option. If so, “export” the file in ASCII, transfer with Mac-2-DOS using the ASCII (text) conversion option, and finally “import” the

file in ASCII format on the destination machine.

Graphic/Art/Paint

Two graphic file standards exist on the Mac: MacPaint format for bitmapped graphics (which can be converted to Amiga IFF) and MacDraw (PICT) format for object-oriented graphics (which cannot easily be converted to IFF or anything else on the Amiga). Of course, EPS is another emerging standard. For the Amiga, fortunately, IFF is a strong standard. Most all Amiga graphics programs import/export IFF files.

Mac-2-DOS offers direct conversion of MacPaint files to/from IFF ILBM format. Just select the MacPaint-IFF conversion option. Because of screen resolution differences between the two machines and the fact that MacPaint format *does not* support color, some image conversions look better than others.

Mac-2-DOS does support EPS files, but *does not* currently convert the bitmap image portion of the EPS file. For further information on PS and EPS, see Desktop Publishing.

Computer Aided Design

AutoCAD Mac is not a widely-accepted standard on the Mac. However, there exist several translators which convert AutoCAD (DXF) files to and from various Mac CAD file formats, such as IGES. There is no CAD file standard on the Amiga. Possibly some compatibility between the Mac/Amiga CAD programs might exist for those programs which support MacPaint/IFF import/export or DXF files (see ACAD Translator).

Clip Art

On the Mac, clip art is available in two formats: MacPaint format and EPS format, with EPS being the newer form. MacPaint clip art can be transferred to the Amiga using the Mac-2-DOS MacPaint-IFF conversion. At this time Mac clip art in EPS format does not appear to be useful on the Amiga. All Amiga clip art seems to be stored in IFF ILBM format, which you can transfer to MacPaint format, if you wish.

Music

In the world of computer music, Standard Midi File (SMF) is the emerging standard for storing musical sequences on all personal computers. More and more programs for both the Mac and the Amiga are SMF compatible. SMF files are directly compatible between the two machines using the “No Conversion”

Mac-2-DOS option.

SMUS is a form of IFF used to store musical sequences by many music programs on the Amiga. SMUS on the Amiga is NOT SMF-compatible; however, SMUS to SMF conversion programs are available for the Amiga. One such program is MIDI Synergy I, from Geodesic Publications. If your Amiga file is in SMUS format, you must perform this SMUS-SMF conversion before the Mac-2-DOS transfer.

Fonts

Unfortunately, font files for the Mac and the Amiga are not directly compatible. However, there is a public domain program for the Amiga, called MacFont, which converts Mac fonts to Amiga fonts. This program can be found on the "Fish Disk" collection of public domain software, and from various BBSs and users groups.

Miscellaneous File Conversion Programs

The products listed in this section provide some form of file conversion which you may find useful in transferring files between the Mac and the Amiga. We have *not* used these programs, so we cannot guarantee that they do what the maker claims. This information is provided merely to let you know that such products exist. Contact the manufacturer for more information.

Mac Utilities

AnyGraph/AnyText - Universal IBM/Mac bitmapped graphics/text translators used with Apple File Exchange. AnyText provides text translation from any PC application into MacWrite file format, while maintaining page formatting information. AnyGraph translates bitmapped graphics from any PC application to MacPaint format. (Compatible Systems Corp.)

CADMover - CAD/CAM graphics file translation program. (Kandu Software)

FontShare - Downloadable PostScript font printing utility. (Olduvai Corp)

MacLinkPlus/Translators - A library of file translation software that adds more than 50 file translation or conversion options (including MacWrite, Word, Word Perfect, OfficeWriter, WordStar, Multimate, XYWrite, etc.) to the Apple File Exchange (AFE) utility. (DataViz)

MGMS IGES/DXF - Transfers data between CAD systems. (Micro CAD/CAM systems)

Postility 1.0 - Provides “A PostScript development environment that also enables the user to manipulate EPS files.” (Postcraft International)

PowerDraw DXF Translator - Translates between PowerDraw CAD format and AutoCAD DXF format. (Engineered Software)

PS VIEW - Allows you to view graphics from PostScript commands and files on screen before sending them to a PostScript printer. (ImageSoft)

The Curator - Conversion utility for MacPaint, TIFF, Glue, PICT, Mac EPS, IBM EPSF, and PostScript formats. (Solutions International)

Amiga Utilities

ACAD Translator - translates DXF (AutoCAD) files to Sculpt/Animate format. (Access Technologies)

3D OPTIONS - Converts IFF bitmapped images to 2D and 3D objects. Supports Videoscape 3D, Aegis Draw, Introcad, Mcad, Professional Page, PostScript, and DXF (AutoCAD). (Rainbows Edge Productions)

Butcher - Graphics utility program for doing such things as changing resolutions, sorting colors, etc. (Eagle Tree Software)

Font Works - Captures font images from IFF files to create custom fonts. (ACS Software)

Interchange - Modular system for sharing objects between 3-D programs. (Syndesis)

MIDI Synergy I - IFF SMUS to SMF conversion. (Geodesic Publications)

PIXmate - Graphic image enhancement utility. (Progressive Peripherals)

Mac Disk Menu

The Mac Disk menu lists special functions which apply only to the Mac disk drive and the disk loaded there. These functions can only be performed if there is a disk loaded in the Mac drive.

Eject Disk

Most Mac disk drives are not equipped with a disk ejection button. This menu item directs the drive to eject the disk. If your drive has an ejection button you do not need to use this function, although it works with all Mac drives.

You can also eject the Mac disk by clicking on the Mac disk volume name above the Mac directory.

Rename Disk

This menu item makes it possible for you to rename the disk volume. When you select this menu item, Mac-2-DOS displays the volume name of the disk and waits for you to enter a new name. Press RETURN when you have entered the new name.

Verify Disk

When you select this menu item, Mac-2-DOS reads all sectors and tracks of the Mac disk to ensure its readability. If the test succeeds, there is no message. But if the test fails, Mac-2-DOS warns you in a requester.

Format Disk

This menu choice lets you create an empty disk in Mac format. You can use any normal 3.5-inch 80-track diskette (such as used on the Amiga or the Macintosh) as a Mac disk. Just load the disk into the drive. Then select the Format Disk option from the Mac Disk menu. Enter the new volume name you wish to assign to that disk. To start the format process, click on One-Sided (400k) or Two-Sided (800k). Normally you should specify the double-sided format, since it has more capacity. However, if your Mac drive is single-sided or if the Mac where you are taking your files uses a single-sided drive, then you should specify one-sided.

Warning: The format process completely erases all previous data of any type which was already on the disk.

File Utilities Menu

The File Utilities menu lists special functions which apply to any selected file, no matter which window the file is in. If no file is selected in either window, these menu choices produce a warning, but perform no action. If you have selected more than one file, these menu choices apply to all selected files, one at a time.

File Info

This menu item displays current information about a file, including:

- Name
- size (bytes)
- creation date (Mac only)
- modification date
- file type (Mac only)
- file creator (Mac only)
- file attributes

This display is “read only”; no changes are permitted to these file parameters.

Display Text

This menu item displays the contents of a text file on the screen. Note that Mac-2-DOS attempts to display the contents of the file as text, whether it is text or not. For some types of non-text files this can produce rather bizarre results.

Rename File

This menu item displays the name of the file and lets you change it. This can be useful where the file name must be changed to make the file acceptable to a particular application program.

Delete File

This menu item lets you delete files you no longer want. Note: if the file is marked as “protected”, Mac-2-DOS is unable to delete the file, and warns you of this fact.

Other Options Menu

The items in this menu are parameters which control the transfer of files. By setting these parameters, you can tailor Mac-2-DOS to meet your requirements. If you change any of these parameters, the change applies only until you exit Mac-2-DOS. If you wish to make the change permanent, you must select Save Options from the Project Menu.

Amiga Icon

If this item is checked (yes), then Mac-2-DOS generates an icon file for every Mac file transferred to the Amiga. The icon file is a supplemental file having the same name as the original file with the addition of “.info”. If this item is not checked, no icon file is generated for the file, and the file will not show up on the Workbench screen.

Ask Amiga Tool

If this item is checked (yes), then Mac-2-DOS asks what default tool name should be assigned to a Mac file which is transferred to the Amiga. If this item is not checked, Mac-2-DOS uses the default tool name predefined for the particular conversion to be applied to the file. See the section entitled **About Amiga Files**.

Ask Mac Type

If this item is checked (yes), then Mac-2-DOS asks what file type and creator codes should be assigned to an Amiga file which is transferred to the Mac. If this item is not checked, Mac-2-DOS uses the predefined codes for the particular conversion to be applied to the file. See the section entitled **About Mac Files**.

Auto-replace

When Mac-2-DOS transfers a file in either direction, it always checks for an existing file of the same name on the destination device before transferring the file. If it finds that the file already exists, it warns you in a requester that the file already exists, and waits for you to decide whether or not to replace the file. If you know that you always want to replace these existing files, you can direct Mac-2-DOS to proceed without asking for your permission (auto-replace).

If this menu item is checked (yes), then Mac-2-DOS automatically deletes an existing file without asking your permission. If this item is not checked, Mac-2-DOS warns you of each existing file and waits for your decision.

Saving and Loading Options

Mac-2-DOS offers user-definable options (see File Conversion Options and Other Options) which allow you to tailor Mac-2-DOS to your requirements. If you decide to change any of these options, the changes apply only until you exit Mac-2-DOS. However, if you wish these changes to become permanent, you can direct Mac-2-DOS to save all user-controlled options in a special options file. Then, whenever you start Mac-2-DOS, it automatically loads the options from the option file, restoring the options to their state when you saved them.

To save the options, select *Save Options* on the *Project* menu.

If, at any time, you wish to restore the options to their previously-saved state, just select *Load Options* on the *Project* menu.

The options file is named M2D.OPT, and is stored in the S: directory.

Problems and Error Messages

If you are having trouble using Mac-2-DOS we want to help you. But please take the time to read the rest of this manual before you call us for help.

Problems with Mac-2-DOS usually fall into two categories: problems in the actual transfer process (bad disks, faulty hardware, operator errors) and problems related to file usability after it has been successfully transferred (wrong format, no icon, wrong file type/creator or default tool name).

Mac-2-DOS Error Messages

Mac-2-DOS accesses Amiga files via AmigaDOS. Many error messages which you may see originate in AmigaDOS. Such messages are not listed here. Please note the name of the originating process at the top of the error requester. If requester title bar is SYSTEM, then the error is not directly from Mac-2-DOS, and is not listed here.

Most Mac-2-DOS error messages are self-explanatory. Listed here are some Mac-2-DOS errors with a more complete explanation of the problem.

Mac drive not found

The Mac-2-DOS interface adapter is not responding to control signals from the program, or there is no Mac drive attached to the interface. Make sure the interface and drive are properly installed. See Hardware Installation.

Mac disk read error

Mac-2-DOS cannot read the necessary data from the disk in the Mac drive. Perhaps the disk is bad.

Mac disk write error

Mac-2-DOS cannot read data from the disk in the Mac drive during a write operation. Because of the way the Amiga disk control hardware works, Mac-2-DOS must read a full track of data before writing any new data to the track. Perhaps the disk is bad.

Can't delete Amiga file

Usually this happens because the file is “locked” for writes by some other program or because the file is marked as “undeletable” by its own protection bits. If the file was left locked by another program, you may have to reboot your Amiga to remove the spurious lock. If the file protection bits are set wrong, you must change the “not deletable” bit using the CLI PROTECT command.

No current Amiga directory

This can happen if you try to change the Amiga device:path to a device and/or which does not exist, then try to perform operations on Amiga files. Reenter the Amiga device:path in the Amiga volume gadget.

Unable to open Amiga file

If this message appears when you try to transfer a Mac file to the Amiga, it means that either the directory or Amiga volume you are writing into is locked for writes. This can happen if the volume is not validated, or if some other program left the directory “locked”.

Amiga directory error: looping

This indicates that the Amiga directory is corrupted. You should exit Mac-2-DOS and fix the directory problem before attempting to use Mac-2-DOS.

Amiga directory error: garbage

This indicates that the Amiga directory is corrupted. You should exit Mac-2-DOS and fix the directory problem before attempting to use Mac-2-DOS.

Can't delete Mac file

The Mac file is marked as “undeletable” by its own protection bits. You should use the File Info menu item on a Mac to change this condition.

Mac disk is full

The Mac disk can't hold any more files. Insert a different Mac disk or format a new Mac disk.

Mac directory is full

Old-style “flat file” (400k) Mac disks allow only a fixed amount of directory space. You have transferred too many files onto the Mac disk. Insert a different Mac disk or format a new Mac disk.

Unknown Mac disk format

The disk you have loaded into Mac-2-DOS does not contain a standard Macintosh disk format. Load a different Mac disk or format a new Mac disk.

File Usability Problems

When you successfully transfer a file to or from a Mac disk, the fun has just begun! You may still not be able to use the file in the way you originally intended. Since the Mac and the Amiga are somewhat different, let's treat them separately.

Mac File Problems

When you click on the Mac file icon, the Mac program won't run.

Let's say you transfer a file to a Mac disk, then take that disk to a Mac. When you open the disk's icon you see an icon for your file, but when you double-click on that file, you see a "dialog box" (that's the Mac name for a "requester") containing this message:

The file "your file" could not be opened/printed
(the application is busy or missing).

This message means that the Finder (the Mac equivalent of Workbench) could not start up the Mac application program specified by the File Creator field of that file. There are several possible causes for this message:

1. The file creator code placed on the file by Mac-2-DOS is incorrect. You may have told Mac-2-DOS to use the wrong code or you have used the wrong file conversion option.
2. The file creator code is correct, but the application program specified by the code really is not available on the Mac.

In this case you must transfer the files from the Amiga again using the correct file type and creator. Alternatively, you can use a Mac public domain program called ResEdit (Resource Editor) to change the file type and creator of the file on the Mac disk so that the Mac Finder can process it correctly.

When you try to open the file from inside a Mac program, the file doesn't appear in the list of available files.

We have found that setting the file type and creator codes on a Mac file is not always sufficient to make a file acceptable to some Mac programs. One example of this is MacPaint. At this time we don't know what else may be necessary, but we know that MacPaint WILL accept the file if you double-click on the file icon to start MacPaint. So be sure you have set the file type and creator correctly, then transfer the file and start the program by clicking on the file's icon.

When you open the file, it contains garbage and/or the Mac program rejects it.

Probably you have selected the wrong file conversion option. For example, if you want to bring a word processing document from the Amiga to the Mac, you will most often have to convert the document to ASCII text on the Amiga, transfer the file using the ASCII conversion option, and then "import" the file

into the Mac word processing program. If you transfer the file using the No Conversion option, the file will almost certainly be rejected by the Mac word processing program.

Amiga Problems

When you click on the file icon, the Amiga program won't run.

Let's say you transfer a Mac file to the Amiga. When you click on the file's icon, Workbench flashes the screen and puts up a cryptic message:

```
Error while opening xxx: 205
```

This message means that Workbench could not start up the Amiga application program specified by the default tool of the icon file. There are several possible causes for this message:

1. The default tool placed in the icon file by Mac-2-DOS is incorrect. You may have told Mac-2-DOS to use the wrong name or you have used the wrong file conversion option.
2. The default tool named in the icon file is correct, but the application program specified by the name really is not available on the Amiga.

If the default tool name is wrong, you can use the Info menu item on the Workbench menu to change the default tool name so that Workbench can process it correctly. If the application program is not available, change the default tool name to a similar application which is available, or load the diskette containing the necessary application into an Amiga disk drive.

When you try to open the file from inside an Amiga program, the file doesn't appear in the list of available files.

Many Amiga programs use a particular naming convention for all files "created" by that program. Usually, on the Amiga, this means appending a "file type" code to the end of the file name. For example, VIP Professional files always end in ".wks". When you transfer a file to the Amiga from the Mac, Mac-2-DOS retains the original Mac name. However, the Amiga program may not accept the file in that form, and you may be forced to rename the file in order to make it usable on the Amiga.

There are three easy ways to rename a file on the Amiga. From CLI use the RENAME command. From Workbench use the Rename menu item. From Mac-2-DOS use the Rename File menu item on the File Utilities menu.

When you open the file, it contains garbage and/or the Amiga program rejects it.

Probably you have selected the wrong file conversion option. For example, if you want to bring a word processing document from the Mac to the Amiga, you will most often have to convert the document to ASCII text on the Mac, transfer the file using the ASCII conversion option, and then “import” the file into the Amiga word processing program. If you transfer the file using the No Conversion option, the file will almost certainly be rejected by the Amiga word processing program.

Technical Information

The Mac-2-DOS interface must be the last device in a daisy-chain of Amiga disk devices. This is because it provides no connector for other Amiga drives.

Mac-2-DOS has been successfully tested with all widely available Macintosh disk drives, including both the original single-sided Apple drive and the newer double-sided Apple drive, as well as drives made by Cutting-Edge, Mirror, and Laser.

Mac-2-DOS has been successfully tested with AMAX, by ReadySoft, to ensure that the devices can coexist on an Amiga. However, each device has its own Macintosh drive connector. The programs cannot be used at the same time, of course. But if you have only a single Mac drive and wish to use it with both devices, you must physically connect the drive to the Mac drive connector of the AMAX device or Mac-2-DOS interface, depending on which device you wish to use. This can be somewhat inconvenient.

Appendix A

Mac-2-DOS Mountlist Parameters

The information provided here is used only for owners of Mac-2-DOS Package B, which includes a separate device driver, and who wish to perform the installation steps manually.

The following is an example of the MountList entry for MF0:, the Macintosh floppy drive connected to the Amiga via the Mac-2-Dos interface. Using your favorite text editor, enter these parameters into the DEVS:MountList file of your Workbench disk.

Note: the unit number is used as a flag to the driver software. If it is 0, the driver performs an auto-config search to determine the proper unit number. You may, however, specify the exact unit where you have installed the Mac-2-Dos interface (unit 1, 2, or 3). In this case the driver uses the unit number you specify.

The device name, MF0:, is hard-coded into Mac-2-DOS and cannot be changed by the user. Contact CCS if you have a device name conflict.

```
MF0:  Device = macfloppy.device
      Unit = 0
      Flags = 1
      Surfaces = 2
      BlocksPerTrack = 11
      Reserved = 2
      Interleave = 0
      LowCyl = 0
      HighCyl = 79
      Buffers = 5
      Mount = 1
```

Appendix B

Macintosh File Type and Creator Codes

Here are the file type and file creator codes assigned to files produced by some of the most common Macintosh programs. Sometimes a program produces files with different file types, depending upon the format of the file as it is stored. In this case, the creator code is the same, but the file type codes differ. In other cases different versions of the same program produce files having the same file type codes but having slightly different creator codes. Note: these codes are registered with Apple Computer, which maintains the “official” list.

Program Name	Type of file	File Code	Creator
Microsoft Word	document	WDBN	MSWD
MacWrite	document	WORD	MACA
	text	TEXT	MACA
WriteNow	document	nX^d	nX^a
PageMaker 2.0	page layout	PUBF	ALD2
PageMaker 3.0	page layout	ALB3	ALD3
ReadySetGo	page layout	RSIG	MART
WordPerfect	document	WPDC	SSIW
Microsoft Excel	spreadsheet	XLBN	XCEL
	text/SYLK/WKS	TEXT	XCEL
ClickOn Worksheet	spreadsheet	CLWD	CLWI
Crunch	spreadsheet	CALC	SOFD
Microsoft Multiplan	spreadsheet	MPB	PLAN
	SYLK	TEXT	PLAN
FileMaker Plus	database	FMKD	FMKR
Ensemble	integrated	MODL	CXCC
Factfinder 1.1	database	FACT	NARU
Microsoft File	database	ISAM	FILE
FileVision	database	PicB	TSP2
Helix	database	HEAP	HELX
Jazz	database	JDBS	JAZZ
	graphics	JGRF	JAZZ
	spreadsheet	JWKS	JAZZ
	document	JWPD	JAZZ
Omnis3	database	OMSD	OM\$\$
OverVUE	database	DVSH	OVUE
MacPaint	graphics	PNTG	MPNT
MacDraw	graphics	DRWG	MDRW
Cricket Draw	graphics	CKDT	CRDW
Full Paint	graphics	PNTG	PANT
Graphic Works	graphics	CBOK	MMCB
Adobe Illustrator	graphics	TEXT	ARTY

Adobe Illustrator 88	graphics	TEXT	ARTZ
	EPS	EPSF	ARTZ
SuperPaint	graphics	SPTG	SPNT
Easy 3-D	graphics	EZ3D	EZ3C
MacDraft	cad	DRWG	MDFT
VideoWorks II	video	VW2P	MMVW